

Tableau WCAG Guide

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Web Accessibility

Ensuring web accessibility is crucial for creating an inclusive online environment at Upstate. We adhere to the Web Content Accessibility Guidelines (WCAG) 2.1, ensuring our web pages are usable by everyone, including individuals with disabilities.

Web Contents Accessibility Guidelines – is an internationally recognized set of standards developed by the **World Wide Web Consortium (W3C)** under its **Web Accessibility Initiative (WAI)**. The purpose of WCAG is to make web content accessible to people with disabilities, including those with visual, auditory, physical, speech, cognitive, language, learning, and neurological disabilities.

A11y – shorthand for Accessibility

Common Unmet Criteria

Infographics often fail to meet several WCAG level AA Criteria including:

- [1.1.1 Non-Text Content](#)
- [1.4.1 Use of Color](#)
- [1.4.3 Contrast \(Minimum\)](#)
- [1.4.5 Images of Text](#)

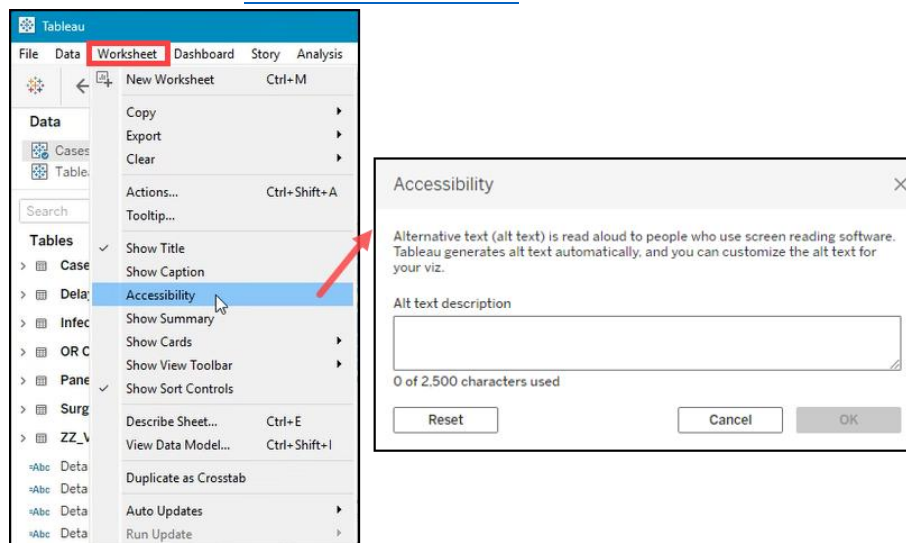
Prioritization of Existing Workbooks

1. Public Facing Workbooks
2. Internal Workbooks where the user is known to need a11y
3. Institute-Wide/Guest Workbooks
4. Highly Utilized Workbooks
5. Remaining workbooks that **are not** Archived

How to Make Compliant Infographics

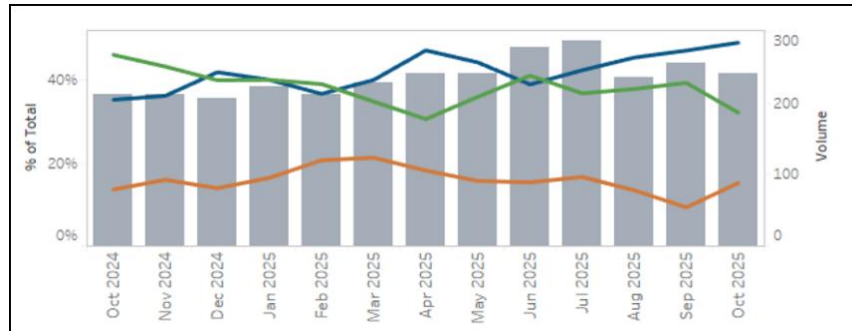
Visualizations

- **No more than one Scroll Bar on a Visual** – Visualizations should not require scrolling in two dimensions unless it is a data details table.
 - Criteria to meet - [1.4.10 Reflow](#)
- **A11y Alt Text** – Each visualization needs an Alt Text description
 - Criteria to meet - [1.1.1 Non-text Content](#)

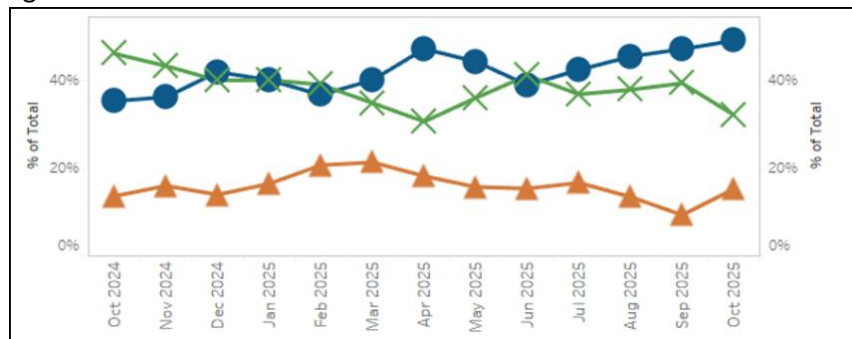


- Have an additional distinguisher other than or in addition to color
 - Criteria to meet - [1.4.1 Use of Color](#), [1.4.3 Contrast \(Minimum\)](#), [1.4.11 Non-text Contrast](#)

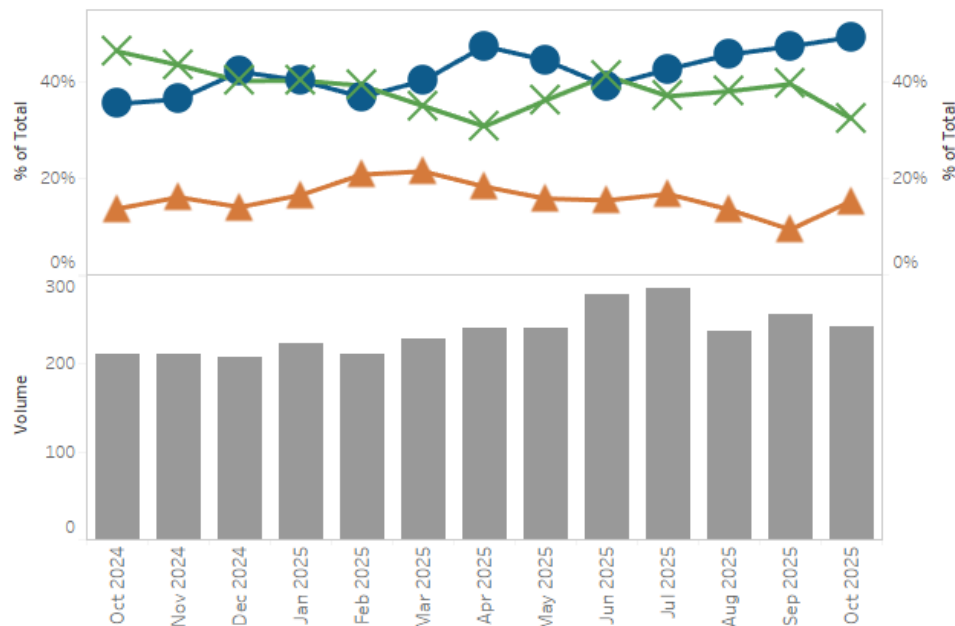
✚ Example of a common visualization that **does not meet criteria**:



✚ To make this visualization compliant shapes can be added to each line to have a distinguisher other than color:

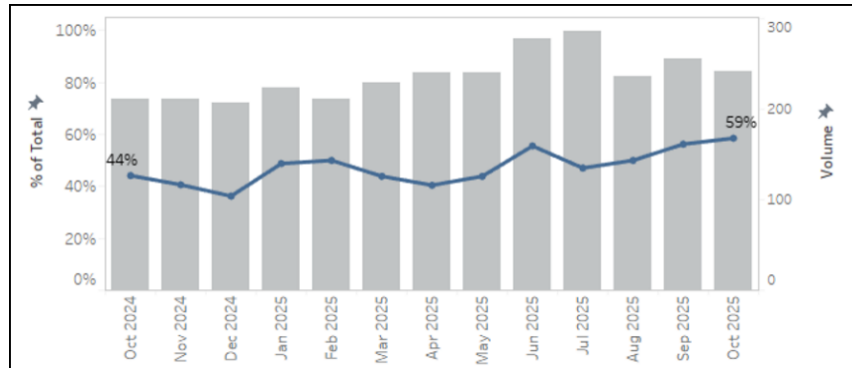


✚ Doing this requires the use of **Dual Axis** and removes the ability to have the lines overlaying bars. To re-incorporate the bars, a second layer can be added that shares the trending months x-axis:



- Use a contrast checker like [WebAIM: Contrast Check](#) - Ensure adjacent colors (back and foreground) have proper ratios.
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#), [1.4.11 Non-text Contrast](#)

✚ An example of a common visualization that **does not meet criteria**.



✚ While the Line:Bar and Line:Background Contrasts meet criteria of 3:1, the Bars:Background is **not compliant**:

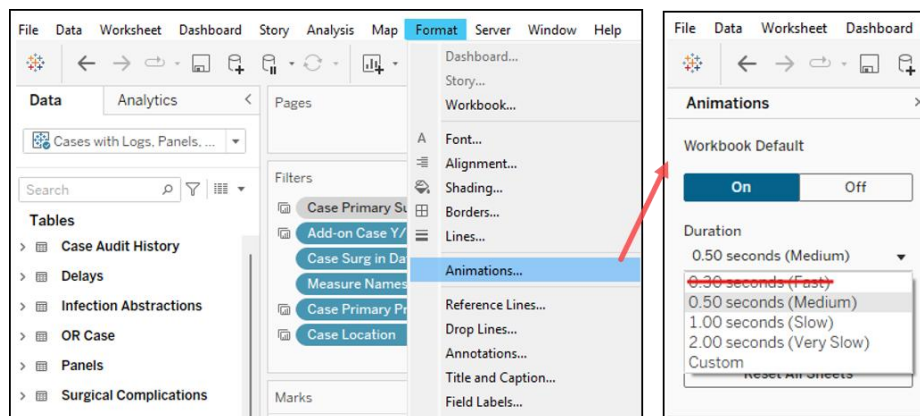
Foreground Hex Value: #466C95 Color Picker Alpha: 1 Lightness Contrast Ratio 3.08:1 Graphical Objects and User Interface Components WCAG AA: Pass	Background Hex Value: #C0C3C4 Color Picker Lightness	Foreground Hex Value: #466C95 Color Picker Alpha: 1 Lightness	Background Hex Value: #FFFFFF Color Picker Lightness Contrast Ratio 5.46:1 Graphical Objects and User Interface Components WCAG AA: Pass
Foreground Hex Value: #C0C3C4 Color Picker Alpha: 1 Lightness Contrast Ratio 1.77:1 Graphical Objects and User Interface Components WCAG AA: Fail			

- ✚ To make this Visualization compliant, dark borders can be added to the bars so that the bar color is no longer adjected to the background for contrast ratio consideration. Consideration is instead given to the Borders with the background and bars.



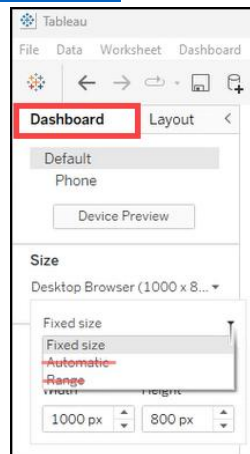
Animations

- Ensure animations do not exceed 3 flashes within 1 second.
 - Criteria to meet – [2.3.1 Three Flashes or Below Threshold](#)



Dashboard

- **Use Fixed Sized Dashboards and Not Automatic Range** – By setting up dashboards with the “Fixed Size” option, zooming in to 200% will not resize the content.
 - Criteria to meet – [1.4.4 Resize Text](#)



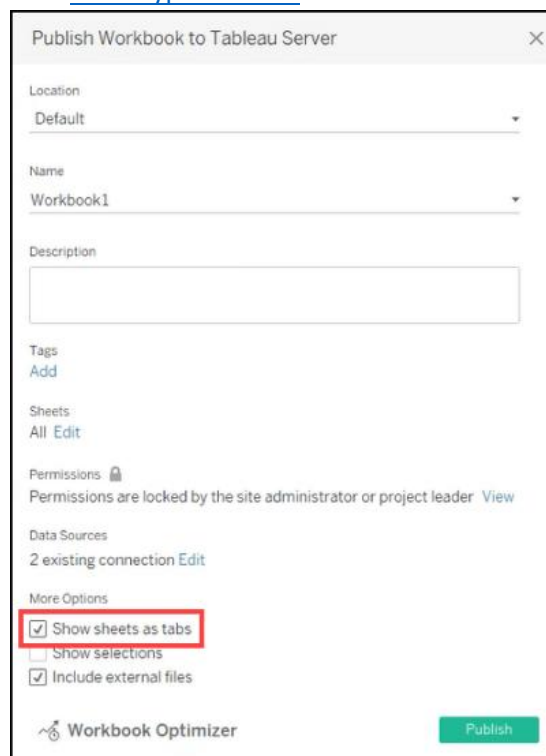
- **Use Text, not images of text** – Does not apply to Logotypes (text that is part of a logo or brand name)
 - Criteria to meet – [1.4.5 Images of Text](#)

Containers

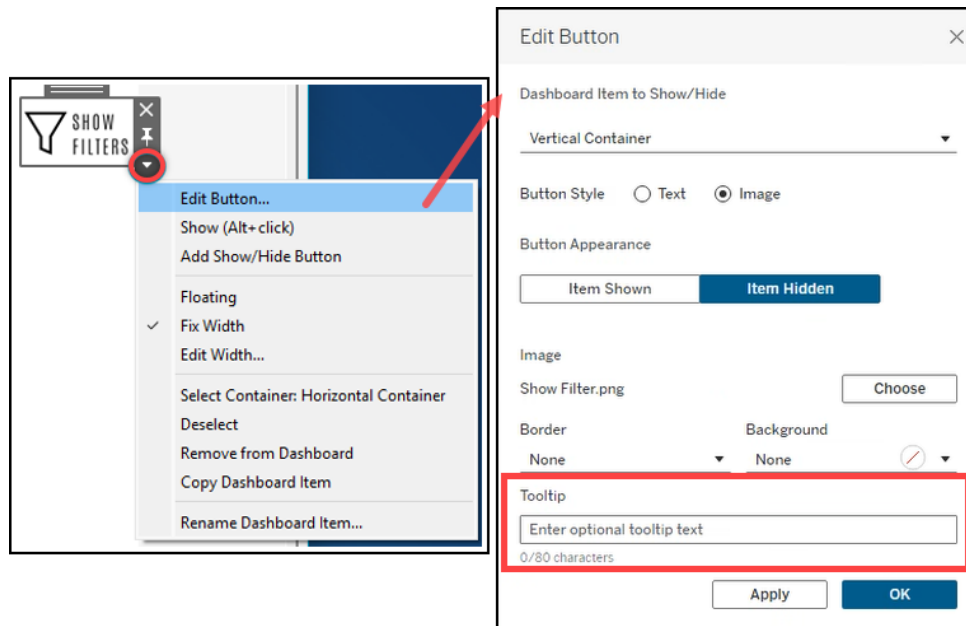
- **Tiled Containers and Flow** – Using Tiled containers instead of Floating will ensure predictable sequencing.
 - Criteria to meet – [1.3.2 Meaningful Sequence](#), [2.4.3 Focus Order](#)

Buttons

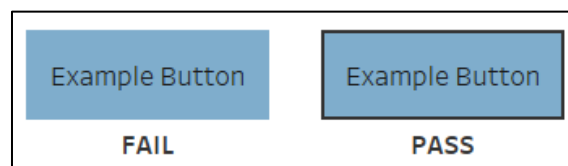
- **Turn on Tabs or have navigation buttons to other Dashboard Sheets.**
 - Criteria to meet – [2.4.1 Bypass Blocks](#)



- **Properly Label and populate Alt Text for Buttons**
 - Criteria to meet – [2.4.4 Link Purpose \(In Context\)](#)

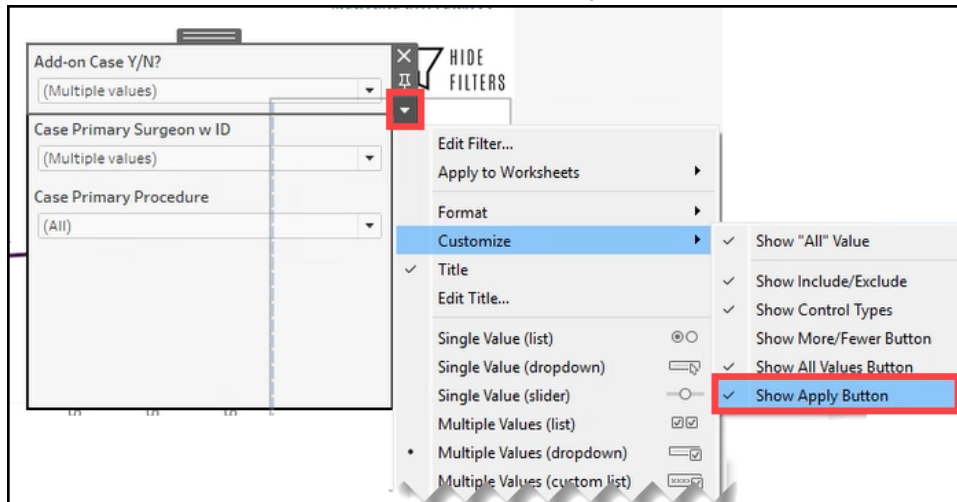


- **Do not use solely shape or position for navigation (ex. An arrow to advance or go back) - Include text to indicate action like 'Next' or 'Submit'**
 - Criteria to meet – [1.3.3 Sensory Characteristics](#)
 - **Consistent placement and appearance of Buttons throughout the Workbook**
 - Criteria to meet – [3.2.3 Consistent Navigation](#), [3.2.4 Consistent Identification](#)
 - **Use a Contrast checker like [WebAIM: Contrast Checker](#) and make sure adjacent colors (background, foreground and text) have proper ratios – Three areas for colors to consider; Dashboard background, Button background, Button Text Adjacent colors need to meet proper ratio requirement of 3:1 and 4.5:1 when normal text is involved. Ensure **Dashboard Background to Button Background** as well as **Button Background to Button Text** meet the appropriate ratio requirements.**
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#), [1.4.11 Non-Text Contrast](#)
- ✚ If it is difficult to get proper ratios, adding a border to the Button may help so there is another adjacent layer of Button Border between the Dashboard Background and the Button Background.



Filters & Parameters

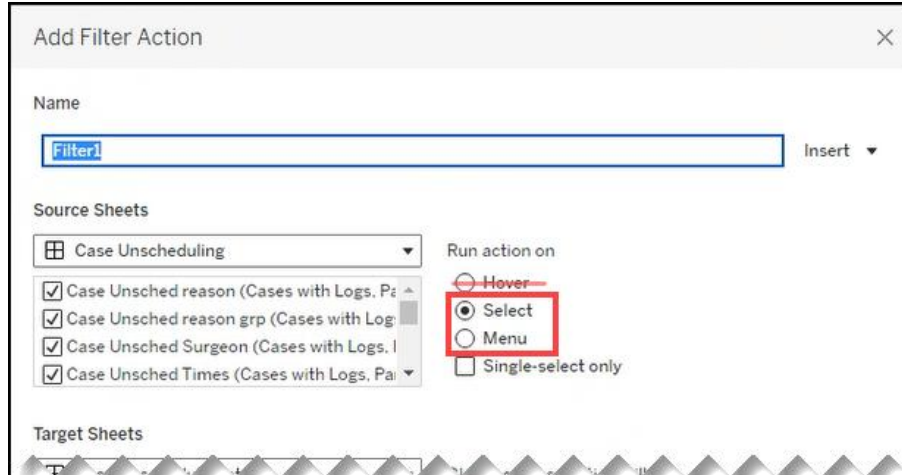
- **Show Apply Buttons or require an 'Enter' press to activate**
 - Criteria to meet – [3.2.1 on Focus](#), [3.2.2 On Input](#)



- **Labels for Filters and Parameters are either Above or To the Left of the entry box.**
 - Criteria to meet – [2.5.3 Label in Name](#)

Actions

- **Filter Actions need to be set to run on Select or Menu, not Hover** – Highlight Action running on Hover is acceptable as content is not changing.
 - Criteria to meet – [1.4.13 Content on Hover or Focus](#)



- **When using menu Action filters, make sure the name is informative of what the action will be.**
 - Criteria to meet – [2.4.4 Link Purpose \(In Context\)](#)

The screenshot shows a dialog box titled "Add Filter Action". It has a "Name" field at the top with the text "Filter to selected Department" inside, which is highlighted by a red rectangular box. To the right of the name field is an "Insert" button with a dropdown arrow. Below the name field is a section labeled "Source Sheets" containing a list of sheets with checkboxes: "Case Unscheduling", "Case Unshed reason (Cases with Logs, P)", "Case Unshed reason grp (Cases with Log", "Case Unshed Surgeon (Cases with Logs, I", and "Case Unshed Times (Cases with Logs, Pai". To the right of this list is a section labeled "Run action on" with four radio button options: "Hover", "Select", "Menu" (which is selected), and "Single-select only". At the bottom of the dialog is a section labeled "Target Sheets" with a list of sheets.

Titles

- **Have Informative Titles**
 - Criteria to meet – [2.4.6 Headings and Labels](#), [2.4.2 Page Titled](#)
- **Ensure adjacent colors (text and background) have proper ratios** – font size 18pt or larger have a ratio of 3:1, smaller than 18pt have a ratio of 4.5:1
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#)

Descriptions

- **Have Informative Dashboard Descriptions**
 - Criteria to meet – [2.4.2 Page Titled](#)
- **Use Text, not images of text**
 - Criteria to meet – [1.4.5 Images of Text](#)
- **Have consistent formatting for text paragraphs and lists**
 - Criteria to meet – [1.3.1 Info and Relationships](#)
- **Ensure adjacent colors (text and background) have proper ratios** – font size 18pt or larger have a ratio of 3:1, smaller than 18pt have a ratio of 4.5:1
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#)

Tooltips

- **Informative Tooltips** – Information in Tooltips should be legible when read on its own containing appropriate context for the Mark. *Example: Clear how percentages are calculated, Values have appropriate labels, etc.*
 - Criteria to meet – [1.1.1 Non-Text Content](#)
- **Have Consistent Formatting for text paragraphs and lists**
 - Criteria to meet – [1.3.1 Info and Relationships](#)
- **Ensure adjacent colors (text and background) have proper ratios** – font size 18pt or larger have a ratio of 3:1, smaller than 18pt have a ratio of 4.5:1
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#)

Formatting

- **Have Consistent Formatting for text paragraphs and lists**
 - Criteria to meet – [1.3.1 Info and Relationships](#)
- **Ensure adjacent colors (text and background) have proper ratios** – font size 18pt or larger have a ratio of 3:1, smaller than 18pt have a ratio of 4.5:1
 - Criteria to meet – [1.4.3 Contrast \(Minimum\)](#)

Post-Publishing

- **Check for Keyboard traps via tabbing through** – Pay particular attention to hide/show containers
 - Criteria to meet – [2.1.2 No Keyboard Trap](#)

Resources

- Upstate Intranet Resources Page- [Click Here to Navigate to Page](#)
- WebAIM Infographic – [Click Here to Navigate to Page](#)
- WebAIM Contrast Checker – [Click Here to Navigate to Page](#)
- WAVE Web Accessibility Evaluation Tools – [Click Here to Navigate to Page](#)
- Official Tableau Help A11y – [Click Here to Navigate to Page](#)
- Official Tableau FAQ: A11y – [Click Here to Navigate to Page](#)
- Official Tableau Help: Build Accessible Dashboards – [Click Here to Navigate to Page](#)